

Widespread Amazonian dark earth in the Xingu Indigenous Territory

Received: 7 November 2023

Accepted: 27 June 2024

Published online: 26 July 2024



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Amazonian dark earth (ADE) is highly nutrient- and carbon-rich soil created by past inhabitants of the Amazon. It would be valuable to know the extent of ADE because of its cultural and environmental importance, but systematic efforts to map its distribution and extent are impractical with traditional field methods. We use remote-sensing imagery and a machine-learning classifier with ground-truthed training data to predict the occurrence of ADE across the 26,000 km² Território Indígena do Xingu (TIX) in the southeastern Amazon region of Brazil. We find widespread ADE across the TIX, well beyond previously studied archaeological sites, occupying at least 3–4% of the land area. We further estimate that the TIX may sequester 9 Mt of carbon within ADE deposits from past human inputs. Our findings show that ancient inhabitants of the TIX substantially modified their environment, highlighting the importance of conserving this natural and cultural resource given threats from climate change and deforestation.

Amazonian dark earth (ADE), also known as terra preta, is a soil type found throughout the Amazon region. It is characterized by dark colour, high organic carbon (C) and charcoal content and elevated soil nutrients compared with the typical extensively weathered and depleted tropical soils found in the region (Fig. 1)^{1,2}. The ubiquitous presence of ceramic fragments and other archaeological artefacts in ADE and the association of ADE deposits with archaeological sites suggests the formation of these soils by human activities; dated ADE deposits typically range from 2500 to 500 years BP (ref. 3). Researchers have argued that the enhanced fertility of ADE was a key factor enabling the development of large and complex prehistoric civilizations in the

region, allowing greater crop productivity and thus population density than natural soils of the region⁴.

In addition to its putative role in supporting past population density, the continued presence of ADE today means that past human activities have sequestered C in the soil for centuries. Soils globally are a major carbon reservoir containing ~3,500 PgC, approximately four times more than the atmosphere and six times more than terrestrial vegetation⁵. Of this global stock, tropical soils contain ~800 PgC, but the long-term stability of tropical soil carbon is threatened both by increasing temperatures⁶ and by land-use change and deforestation^{7,8}. Natural soils of the Amazon are typically low in C (ref. 9), but ADE is

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Fig. 1 | ADE. **a**, Test pit within ADE deposit at Akagahütü archaeological site in TIX, showing characteristic dark colour and ceramic fragments. **b**, Test pit at the periphery of the site, showing red colour typical of extensively weathered

Amazonian soils. **c**, Distribution of ADE deposits known from published studies. Figure adapted with permission from: **a, b**, ref. 31 under a Creative Commons license CC BY-NC 4.0; **c**, ref. 17, Project MUSE.

substantially enriched above natural levels in organic C and additionally contains abundant charcoal^{10,11}. The ADE deposits may therefore represent a substantial carbon reservoir beyond natural soil, sequestered by past human action¹². Moreover, ADE could offer a template for continued carbon sequestration and sustainable tropical agriculture.

Despite the cultural and environmental importance of ADE, the number and size of ADE sites and the overall extent of human soil modification in the Amazon region are unknown. The poor constraints on the spatial extent and abundance of ADE hinder estimates of pre-colonial populations and their locations, prevent an accurate inventory of anthropogenic carbon sequestration, and make the suitability of ADE for widespread sustainable agriculture unclear. Effective conservation of this natural and cultural resource depends on reliable estimates of its distribution.

Known deposits of ADE are typically identified and studied by labour-intensive fieldwork and excavation^{13–16}. While this approach yields valuable information about the detailed composition and spatial structure of ADE at individual archaeological sites, it is not well suited to systematic surveys of large regions due to dense vegetation cover, logistical impracticality and difficulties of access. As a result, reported ADE locations (Fig. 1c) are concentrated in well-studied areas and along major rivers and modern roads that provide access for researchers. Compilations of reported deposits probably omit the majority of all ADE deposits because many regions have not been surveyed for the presence of ADE^{17–20}. Attempts to analyse the spatial distribution or overall extent of ADE from such databases are inherently hampered by the incompleteness of reported data.

Despite these challenges, the locations of known ADE deposits correlate with certain environmental characteristics, suggesting that it may be possible to map ADE over larger areas using remotely measured proxies. One study used a maximum-entropy model to estimate the probability of finding ADE across the Amazon region using a database of known locations and a suite of environmental variables, such as topography and climate¹⁸. This approach predicts regions where ADE is likely to be found and identifies key environmental predictors of ADE. This result raises the possibility of using geospatial data to map the location and extent of specific ADE deposits in regions with sufficient ground-truth data.

Satellite remote-sensing datasets feature global coverage at high spatial and temporal resolution and have been used for land-cover classification at large scales^{21,22}. Analysis of remote-sensing imagery may thus offer the opportunity to detect and map specific ADE deposits while reducing the sampling biases that are difficult to avoid in on-the-ground field surveys²³. While typical remote-sensing modalities cannot directly image the soil of forested regions due to dense vegetation and tree canopy cover, forest growing on ADE has observable

differences from forest growing on unmodified soil due to different soil properties, which manifest as a detectable contrast in satellite imagery^{23–25}, and other studies have found differences in bare soil reflectance in deforested areas²⁶.

To identify and delineate specific ADE deposits, we used Landsat 5 imagery to detect and classify upland forest growing on ADE in the 26,000 km² Território Indígena do Xingu (Xingu Indigenous Territory) (TIX), located in the southeastern Amazon basin (Fig. 2). The TIX preserves a fragment of intact forest under Indigenous management within the southern Amazonian ‘arc of deforestation’²⁷. Fieldwork in the TIX has yielded valuable information about ADE at a local scale^{28–30}; we used remote sensing to upscale these findings to the entire study region. We applied two machine-learning classifiers to complementary training datasets assembled from field observations and manually labelled imagery (Methods), representing ADE with distinct land-use histories (Fig. 3): the first for areas that have experienced twentieth-century deforestation and subsequent reforestation and the second for areas that have probably been undisturbed for several centuries. We used the two resulting classification maps, filtered to reduce pixel-scale noise, to identify and delineate specific ADE deposits, assess the similarity of ADE deposits across the region to the well-studied sites near the Kuikuro village and analyse the spatial distribution and overall areal abundance of ADE and its varied land-use history within the TIX.

Results

Accuracy of classifier

We find that our classification and filtering algorithm reproduces the land-cover data with high accuracy, which we assessed by partitioning the data into training and evaluation sets (Methods); we do this separately for each of the two classifiers and datasets. For the secondary-forest training data, we observe that the classifier correctly identifies 96.9% of forest on unmodified soils, misclassifying 3.1% as ADE. Since ADE is the variable of interest, we consider the ‘unmodified’ pixels incorrectly classified as ADE to be false positives, with a 3.1% false-positive rate (Supplementary Table 1). For known ADE, we observe that the classifier correctly identifies 94.7% of pixels (true positives, green shading), misidentifying the rest as unmodified soil (second row, first column); we consider this error to be a 5.3% false negative rate (blue shading). For the undisturbed-forest training data, we find slightly higher accuracy, with 0.4% false-positive rate and 3.9% false negative rate (Supplementary Table 2).

Classification of the TIX

Our two filtered classification maps predict ADE throughout the TIX uplands, although with different patterns for each (Fig. 4). The

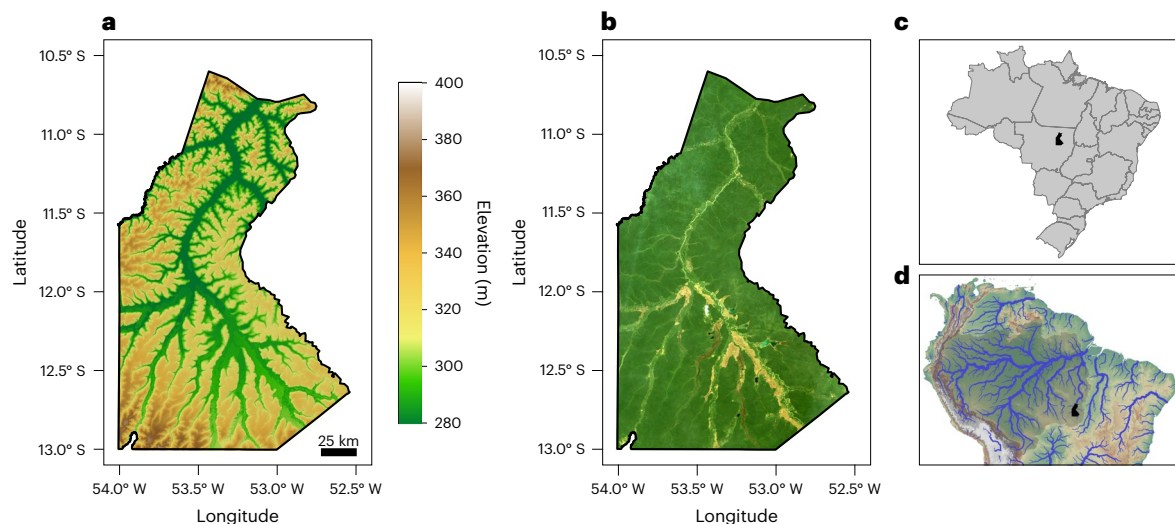


Fig. 2 | TIX, Mato Grosso, Brazil. **a**, Colour-shaded topography of TIX. **b**, True-colour RGB Landsat 5 image mosaic of TIX. **c**, Political context of TIX (black polygon) in Brazil. **d**, Physical context of TIX. Blue lines show major rivers; shaded region marks the extent of the Amazon River basin.

classification map using the secondary-forest dataset predicts 1,850 km² of ADE across the TIX, largely in the south and east of the region. These include archaeological sites studied during 2019–2021, as well as previously unidentified deposits³¹. The classification map also exhibits striping artefacts perpendicular to the direction of Landsat orbit (Fig. 4b, upper right). By contrast, the ADE locations predicted using the undisturbed-forest training data are found primarily in the west and north of the TIX, with an extent of 910 km². In both, predicted ADE locations are dominantly discrete deposits with a rounded planform geometry, typically found near floodplains and frequently elongated in the direction of the bluff (floodplain-upland boundary) (Fig. 4d). The rest of the classified upland area is classified mostly as forest on unmodified soil (dark green), with a few areas of open ground (yellow).

Discussion

Comparison of the two datasets

Our two training datasets both suggest extensive ADE, although they yield different results in several ways due to their different vegetation signatures and land-use histories (Fig. 5). The classifier gives higher accuracy using the undisturbed-forest dataset than using the secondary-forest dataset. This is unsurprising since the ADE polygons in the undisturbed-forest dataset were chosen for their distinct characteristics in remote-sensing imagery, rather than from direct observations of the soil. As a result, the ADE training data are more similar in their spectral signature. The results using the secondary-forest data predict 3.5 times more ADE in the TIX than using the undisturbed-forest dataset. This may be partly due to the higher false-positive rate. The classification map using the secondary-forest data shows large swaths of predicted ADE in the east that may not be realistic and further shows apparent artefacts from the satellite imagery, such as striping and edge effects along Landsat tile boundaries.

Other than these obvious artefacts, we do not believe that the classification results based on either type of training data are inherently more correct than the other. Rather, they each detect deposits of ancient ADE that differ from one another in recent land-use history. Most of the ADE sites used for the secondary-forest data have been recently cultivated (within the past century) by Kuikuro farmers³². However, their distinct spectral signature indicates that the soil retains properties distinct from surrounding areas despite recent human impact. Conversely, the undisturbed-forest polygons elsewhere in the TIX probably represent regions that were depopulated and abandoned following colonization and have not experienced substantial human

land use since abandonment. As a result of these land-use differences and resulting vegetation differences, the two types of ADE deposits show distinct remote-sensing signatures. The secondary-forest ADE deposits near the Kuikuro village are noticeably lighter than surrounding forest in the visible bands but darker in the near-infrared (NIR). The undisturbed-forest ADE deposits elsewhere are not substantially different from the surrounding forest in the visible wavelengths but are markedly brighter in the NIR and, to a lesser extent, short-wave infrared (band 5), indicating higher primary productivity and vegetation richness³³. As a result of these differences, the classification maps produced with each type of training data predict different patterns of ADE (Fig. 5). Beyond the apparent noise/false positives in the east and northeast of the TIX, the map using secondary-forest data largely predicts ADE deposits near the Kuikuro village and the nearby archaeological sites used for training data. Conversely, the map using undisturbed-forest polygons largely omits the known archaeological sites near the Kuikuro village that have been recently cultivated but identifies many sites in the north and west of the TIX not identified by the secondary-forest data.

Thus, the small fraction of mutual agreement (220 km²) between the ADE deposits predicted by the two training datasets (Fig. 5) is probably not an indication that either prediction is erroneous, nor is the area of overlap necessarily a good proxy for the terrain most likely to contain ADE. Instead, the two training datasets probably represent complementary approaches that identify ADE deposits with distinct land-use histories. A comprehensive analysis should include both sets of results to account for the diversity of present-day ADE deposits. The union of the two sets of predicted ADE locations shows that ADE is found throughout the TIX but that there are geographic trends in the twentieth-century cultivation history of ADE deposits. This result indicates that ancient inhabitants of the TIX were probably widespread across the region and had a wide-ranging environmental impact, rather than occupying only a few isolated settlements.

Spatial patterns and relationships with landscape structure

Similarly to the well-studied sites in the Kuikuro territory^{30,31,34}, both classification maps predict that ADE deposits are located primarily along the bluff edges, near the slope down to the floodplain (Figs. 4b,d and 5). These areas typically feature similar topography to adjacent bluff areas without predicted ADE, indicating that the detected differences in vegetation are not the result of topographic differences. Many predicted ADE sites are found where a small tributary joins the floodplain of a larger river, in some cases spanning both sides of the

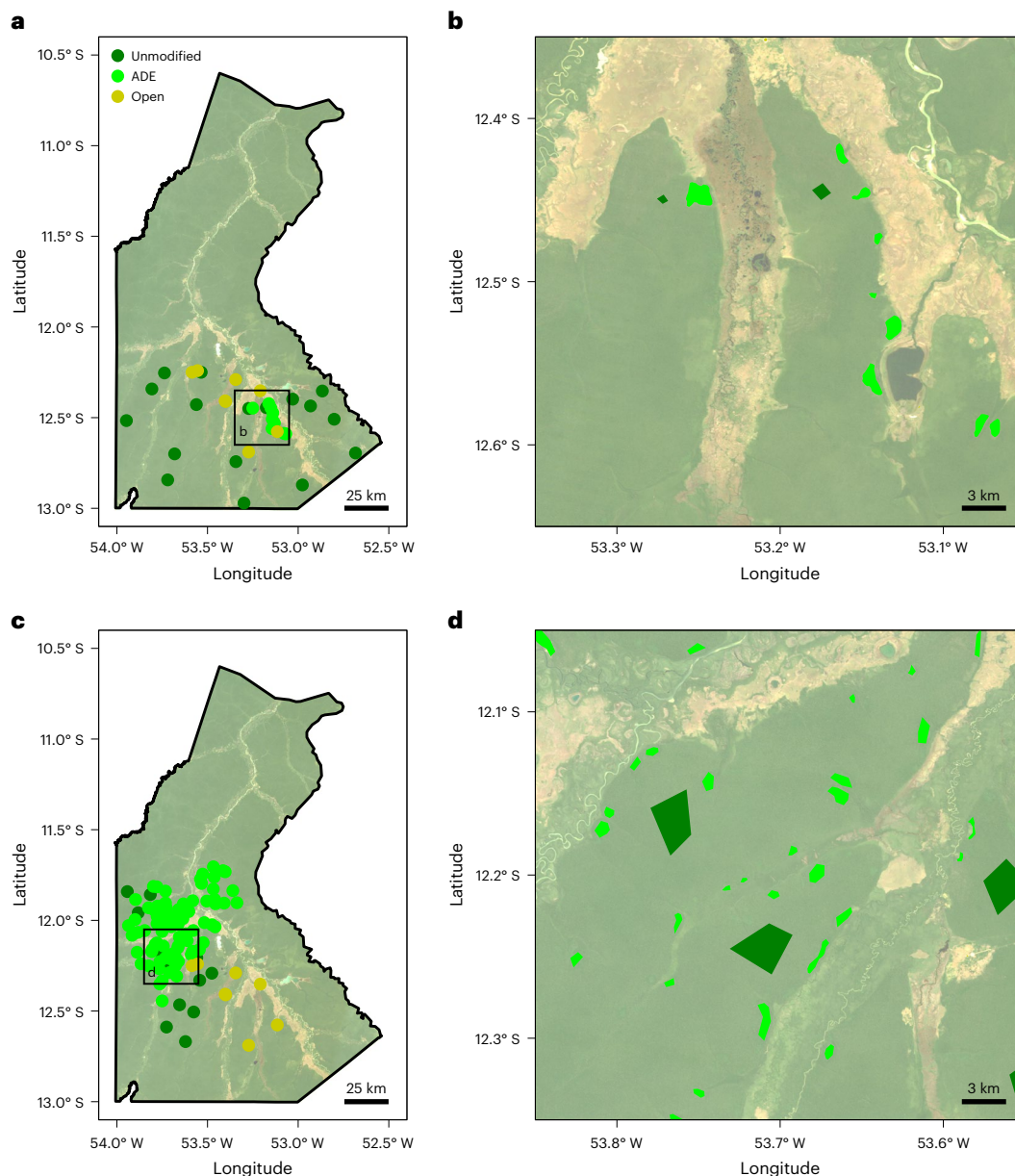


Fig. 3 | Training data. **a**, Training data of ADE under secondary forest. **b**, Close-up of **a** showing outlines of ADE locations. **c**, Training data of ADE under undisturbed forest. **d**, Close-up of **c**.

tributary. The predicted deposits are generally 50–200 ha in size, which is comparable to the size of the well-studied sites³⁰. This pattern is consistent throughout the TIX and strongly suggests an environmental preference for ancient settlements near the interface of the uplands and the floodplains. Our predictions also show that ADE sites are typically spaced 2–10 km apart along the bluffs, with intervening areas of unmodified soils in otherwise environmentally similar areas.

A similar pattern has been observed for archaeological sites throughout Amazonia³⁵. Reference 18 used a maximum-entropy model based on a database of known ADE locations across the Amazon and environmental data such as topography and climate to identify regions likely to contain ADE. They found that elevation and distance from the nearest bluff edge were among the most important predictor variables, with ADE most likely to be found near bluff edges. Our analysis, which uses surface reflectance data that directly result from soil properties rather than environmental or topographic characteristics, finds the same trend among the specific deposits we identify: the ADE sites of the TIX, even those unmapped and unstudied, are almost universally found

along bluff edges. The strong agreement between these two different approaches illustrates a clear geographic preference among ancient Amazonians in the upper Xingu for ADE formation on river bluffs and supports the findings of ref. 18, which predicts a high likelihood of ADE along the bluffs where we identify ADE deposits. Our complementary approach predicts their specific locations, size and spacing, allowing us to evaluate the inventory of ADE in specific areas.

Spatial extent and implications for carbon storage

Past research has demonstrated organic carbon densities in the upper 1 m of soil of 10–25 kg m⁻² for ADE^{10,31}. In addition to the organic content, charcoal may make up 30–50% of the carbon in these soils¹¹. For comparison, typical latossolos of the region have average carbon densities of 7.5 kg m⁻² (ref. 9). If we take as a coarse approximation that ADE on average holds 10 kg m⁻² of additional anthropogenic carbon above the natural abundance, then a 100 ha ADE deposit (a size typical of the studied and predicted sites) contains ~10,000 tonnes of anthropogenic C. From our filtered classification image using the undisturbed-forest

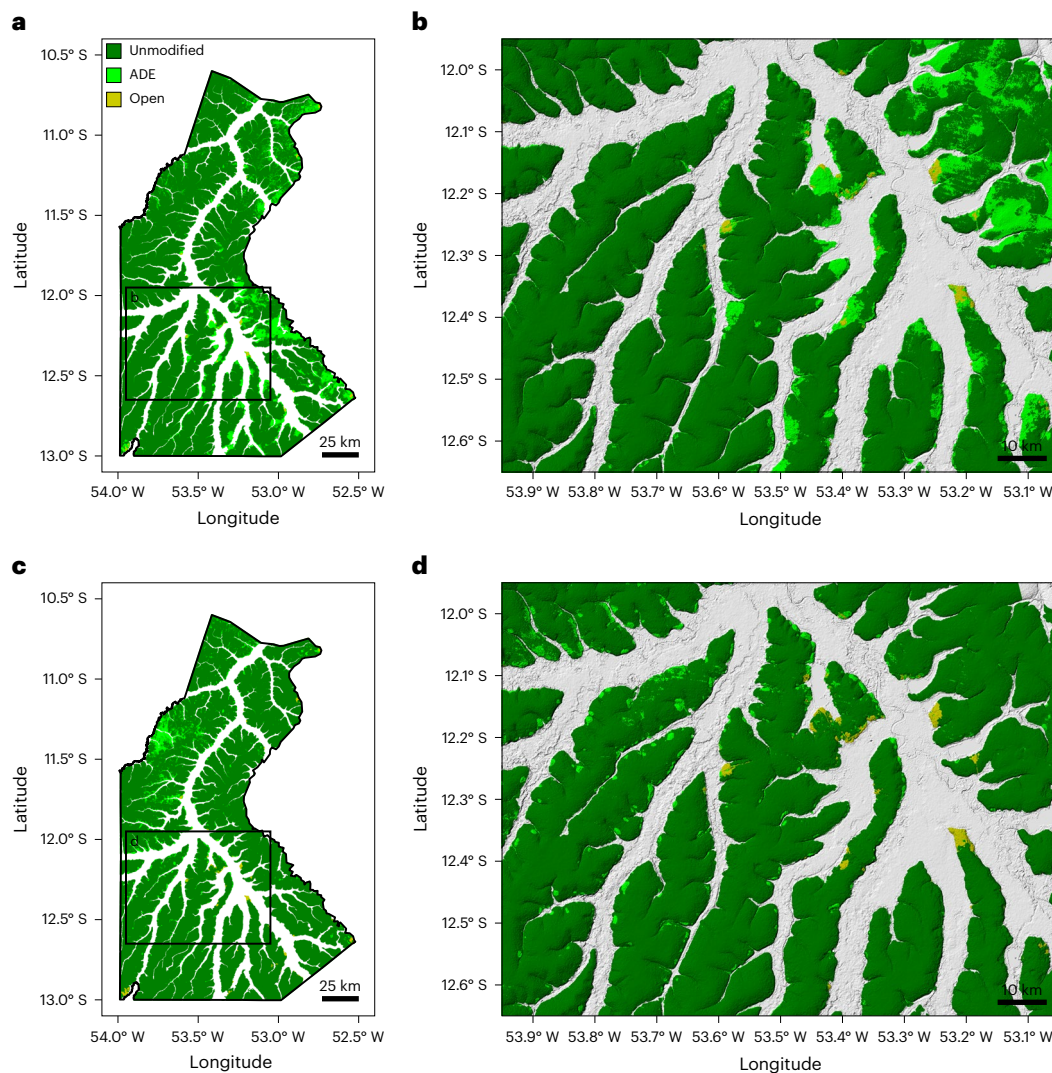


Fig. 4 | Classification maps from random-forest classifier. **a–d**, Results using training data based on sites under secondary forest (**a,b**) and those that are probably undisturbed (**c,d**). Panels **b,d** show close-ups of **a,c**, respectively,

with relief shading showing the topography (using MERIT DEM⁴⁶). The white uncoloured areas within the TIX are the floodplain areas excluded from the analysis, as described in Methods.

data (Fig. 4c), we estimate 910 km² of ADE within the TIX. The TIX is 26,500 km² in total with 20,000 km² of uplands, meaning that 3.4% of the total area and 4.6% of the upland area is composed of ADE. This fraction is consistent with a previous study applying a similar approach to deforested land elsewhere in Amazonia²⁶. Applying the same estimate of ADE carbon abundance gives ~9 Mt of anthropogenic C sequestered in ADE in the TIX, comparable to the annual CO₂ emissions of New Zealand³⁶. This estimate represents a more conservative assessment of the areal extent of ADE than the results using the secondary-forest data (Fig. 4a), which may overpredict the abundance due to imagery artefacts. The low false-positive rate of the undisturbed-forest classifier means that this estimate is probably reasonably reliable although it omits ADE deposits that have experienced recent cultivation, meaning that the true areal extent of ADE in the TIX is probably higher than 3.4%. Even considering the uncertainties in these quantitative estimates of carbon sequestration, our classification maps show extensive and widespread anthropic modification of soils and the environment throughout the TIX.

Implications for Amazonia

Our remote-sensing analysis focuses on the TIX because it is forested and under Indigenous management, in contrast to the extensive deforestation and agriculture in the surrounding area. While our approach

cannot easily be extended to adjacent deforested cropland, it is likely that such areas feature similar concentrations of ADE as the TIX; the environmental imprint of past human activities is not limited by modern boundaries, and predicted ADE is widespread up to the boundaries of the present-day TIX. The environmental degradation of adjacent unprotected lands highlights the importance of conservation for the TIX and other Indigenous areas throughout the Amazon to preserve the natural and cultural resource of extensive ADE deposits.

The TIX represents less than 0.5% of the 5,500,000 km² Amazon rainforest. Although our analysis does not quantify the ADE fraction for the larger region, if the entire Amazon has a comparable fraction of ADE deposits to the TIX, ADE could represent a globally substantial carbon reservoir. However, there is some uncertainty as to whether it is reasonable to extrapolate values for the TIX to the entire Amazon. While the upper Xingu region was densely occupied in the past with major human alteration of the landscape, it is unclear whether such dense habitation occurred throughout the Amazon. Evidence suggests that some regions, particularly along major rivers, were more densely occupied and for a longer duration than the upper Xingu basin and feature substantially more extensive and fertile ADE³⁵, but other regions such as interfluvial areas were probably more sparsely settled. Nevertheless, ADE is widespread throughout the region^{17,18}. Reference 18

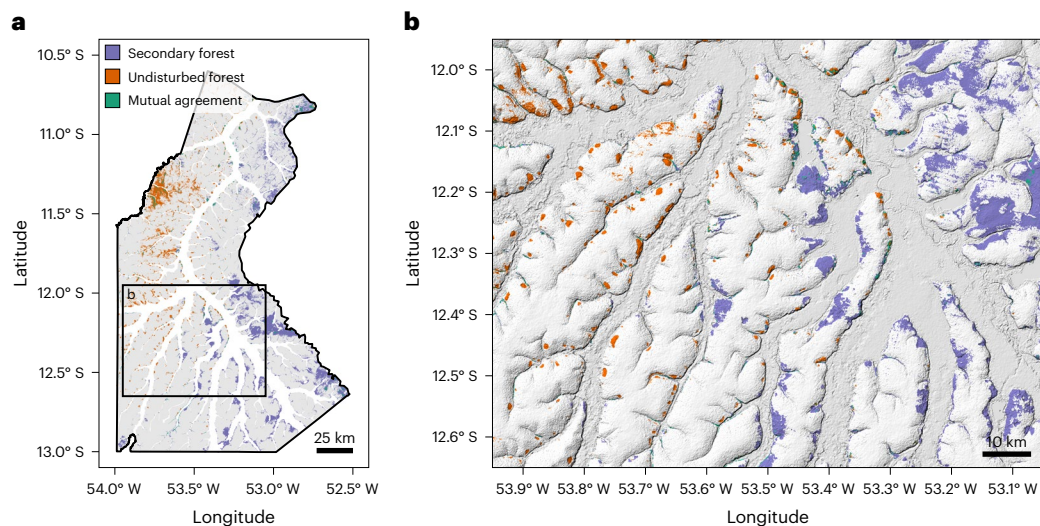


Fig. 5 | Comparison of the two classification maps. a, Consensus map of each classification map shown in Fig. 4. Blue areas denote ADE predicted when using the secondary-forest training data, and orange areas denote ADE predicted

when using the undisturbed-forest training data. Green areas denote mutual agreement of the two results (areas predicted as ADE by both). **b**, Close-up of **a** with relief shading showing topography as in Fig. 4.

produced a likelihood map for finding ADE throughout the Amazon, and similarly predicted a high likelihood in many, although not all, parts of the region. ADE thus probably represents a substantial carbon sink beyond the carbon naturally present in Amazon soils. Moreover, the practices used to create ADE in the past could be applied in the present and future to sequester additional carbon in tropical soils³¹.

Future directions

With a few exceptions, the bright patches in the NIR that we use in our undisturbed-forest dataset do not have confirmed ADE from field investigation. As a result, our use of these features as putative ADE deposits is speculation. However, given their distinct signature and clear distribution, it would be straightforward to investigate the soil properties of these locations to test this hypothesis. If these sites are confirmed to be ADE deposits, then the classification map produced using the undisturbed-forest training data is probably the most reliable estimate of the locations of ADE in the TIX.

While this study demonstrates the potential usefulness of remote sensing and machine learning for ADE identification within a region, the TIX is only a small part of the Amazon rainforest. The upper Xingu basin is located near the forest–savanna boundary in the southern Amazon³⁷. This transitional ecosystem may be particularly sensitive to changes in soil properties or moisture, such as those that result from ADE, making the TIX an ideal test case for the detection of ADE through remote sensing of vegetation. Due to regional variations in vegetation and forest ecology, a single trained classifier may not be applicable over the entire region, and a classification algorithm must be trained on data within the region of interest to be analysed. Therefore, replicating this analysis in other regions must necessarily incorporate local field data. The satellite archaeology approach we employ is not a replacement for field investigation but rather a way to upscale field observations in ways that are not practical or possible with traditional methods. When applied to locally relevant training data, the machine-learning and remote-sensing approach employed here can potentially illuminate the hidden extent of ADE in other regions across the Amazon.

ADE is a key part of the Amazonian archaeological record, an important aspect of indigenous cultural heritage, a major contributor to biodiversity and sustainable land use and potentially a major carbon reservoir, but it is threatened by land-use change and climate change. We demonstrate the applicability of ADE detection from remote sensing in the TIX and find that ADE occurs consistently across the entire

region, well beyond the areas that have been explored archaeologically. This result demonstrates widespread human impacts on the ecosystem, showing that ADE can be a viable template for sustainable tropical agriculture. Moreover, ADE potentially represents a large carbon sink, as a result of past human action, that has remained sequestered in the soil for centuries. Our findings underscore the potential of collaboration with Indigenous peoples to tackle globally relevant problems, including critical ecosystem services and land conservation, while at the same time benefitting local communities, as the southern Amazon approaches an imminent tipping point^{38–40}.

Methods

Study region

We focused our analysis on the TIX in the upper Xingu River basin in the southeastern Amazon, Mato Grosso State, Brazil (Fig. 2). The geologic substrate of this region is the Tertiary Ronuro Formation, consisting of poorly consolidated terrestrial sand⁴¹. Fluvial and hillslope processes have shaped the landscape into a mosaic of meandering rivers and adjacent floodplains dissecting the terra firme forested uplands, which are characterized by gently curving, parabolic hillslopes with a relief of 30–50 m above the floodplain (Fig. 2a). The region's climate is classified as a tropical savanna, with a mean annual temperature of 25 °C and 1,800 mm yr^{−1} of precipitation concentrated in the wet season between October and April⁴². The TIX is located near the southern boundary of the Amazon rainforest, along the forest–savanna transition. The unprotected region outside the TIX has been extensively deforested for agriculture in recent decades; the TIX preserves a fragment of forest under indigenous management⁴³. However, the forest of the TIX has been extensively affected by twenty-first-century wildfires that have burned much of the region³⁷.

The TIX is inhabited by over 7,000 residents from 16 ethnic groups that comprise several Indigenous peoples, including the Kuikuro nation within the Xinguano culture of the southern TIX. We use training data from the Kuikuro community and their traditional territory, which features well-studied archaeological sites on upland bluffs adjacent to the Culuene (Xingu) and Angahuku (Buriti) rivers^{28–32,34,44}. Archaeological and ethnographic research has documented culturally continuous occupation for centuries up to the present, including radiocarbon dates of ancient sites that range from 2,100 years BP to modern, primarily between 1,000 and 300 years BP. These sites demonstrate a large, complex and densely settled society in the past. Archaeological sites

and ADE deposits are typically under forest cover on bluffs near the floodplain, although some sites have been deforested for small-scale agriculture.

Imagery sources

Due to the recent modification of the forest landscape by fires, we used Landsat 5 satellite imagery from 1985 to 2005, before substantial burning of upland forest. We used the atmospherically corrected surface reflectance dataset from the Enhanced Thematic Mapper sensor, which consists of six reflectance bands in the visible, NIR and short-wave infrared (450–2,350 nm), as well as a brightness temperature band computed from emitted long-wave infrared (10.4–12.5 μ m). To leverage seasonal differences in vegetation, we computed two composite images over the TIX in Google Earth Engine⁴⁵: one for the wet season (1 October–30 March) and one for the dry season (1 May–31 August). We calculated the pointwise median of all cloud-free pixels for each season between 1 January 1985 and 1 January 2005 on a 30 m UTM (Universal Transverse Mercator) grid (zone 22S); the cloud mask was taken from the associated quality attribute layer for each image.

Over the TIX, the median number of cloud-free measurements at each point, which we used to compute the composite image, is 84 for the dry season and 20 for the wet season, with maximums of 219 and 51 and minimums of 77 and 7, respectively. The smaller number of wet-season measurements is the result of higher cloud cover. Since our analysis focuses on the forest uplands, we masked out the floodplains to reduce the overall variation in the image (Fig. 4). We computed a floodplain mask using a topographic threshold of 6 m above the closest channel with drainage area greater than 30 km², using MERIT Hydro topographic and hydrographic data^{46,47}. We combined the two images (wet- and dry-season composite) into a single 14-band image which we used for the classification algorithm. We do not include environmental or topographic data as inputs because we aim to identify specific ADE deposits rather than regions likely to contain ADE; field observations show that nearby areas with identical topographic and environmental characteristics can differ substantially with regard to soil properties and anthropogenic influence, with ADE deposits typically structured and spaced out by several kilometres along the bluff edge^{29,44}.

Training data

Extant ADE deposits within the TIX have diverse land-use histories. Some are presently deforested and actively used for small-scale agriculture, some are covered by secondary forest following recent or historic deforestation and cultivation, and others have probably been undisturbed for centuries. We did not consider the deforested sites in our analysis due to their complex and varied appearance in satellite imagery. However, there are clear differences between the remote-sensing signatures of reforested and undisturbed sites. Secondary forest growing on ADE deposits near the present-day Kuikuro village is noticeably brighter (higher reflectance) than surrounding forest in the visible bands (Landsat bands 1–3), but darker in the NIR (Landsat band 4). Forest growing on undisturbed ADE deposits farther from the modern village is not substantially different from the surrounding forest in the visible wavelengths but is markedly brighter in the NIR, and to a lesser extent in short-wave infrared (Landsat band 5). We note that some past ADE deposits may no longer retain properties of ADE due to subsequent natural or anthropogenic processes. We expect that vegetation in such areas will resemble that on soils that were never modified, as a function of the present-day soil properties. We classify such areas as ‘unmodified’ since they are not ADE in the present day.

To represent these two different remote-sensing signatures of ADE, we used two training datasets of labelled polygons. For each dataset, we defined three land-cover categories: forest on unmodified soil, forest on ADE and open ground (Fig. 3). For the first training dataset, representing ADE under secondary forest (Fig. 3a,b), we derived polygons for the ADE category from field excavations and mapping

of nine studied archaeological sites near the Kuikuro village known to have been deforested and cultivated in recent times³². This category consists of 9 polygons ranging from 13 to 211 ha in area, with an average of 87 ha. Polygons for the other two categories were drawn from satellite imagery on the basis of a qualitative land-cover assessment. Seventeen unmodified soil polygons were drawn in the interior of nearby uplands in areas with no indications of forest disturbance and distant from bluffs where known archaeological sites occur. Although we do not have systematic field survey data across the entire region, these non-ADE areas were chosen on the basis of non-systematic survey over three decades of field research and extensive local knowledge of ADE locations near the Kuikuro village. Eight bare-ground polygons correspond to cleared areas, typically in or adjacent to modern villages.

For the second dataset, representing ADE under undisturbed forest, we instead drew polygons for the ADE class around distinct bright patches of forest in the NIR (Landsat band 4). These deposits typically feature band 4 surface reflectance values of 0.30–0.34, in contrast to 0.26–0.29 for adjacent forest (Supplementary Fig. 1). The increased NIR surface reflectance indicates higher primary productivity and vegetation richness³³ and has been associated with ADE deposits elsewhere in the Amazon²⁵. While we do not have direct field evidence for ADE in most of these patches, several are associated with known archaeological sites. Moreover, their spatial pattern, typical size and location relative to topographic features are consistent with characteristics of known ADE sites elsewhere in the TIX. For example, many of the NIR-bright patches are located along the edges of bluffs or form pairs of equal-sized patches on opposite sides of small tributary streams (Supplementary Fig. 1). We manually drew polygons around 119 of these patches ranging in area from 7 to 132 ha (average 42 ha) and 12 areas of nearby unaltered forest (average 980 ha). We used the same polygons for open ground as in the first dataset.

Classification algorithms

We used two random-forest classifiers⁴⁸, one for each training dataset, implemented in Google Earth Engine⁴⁵. We used 100 trees for each classifier instance, a bag fraction of 0.5 and $\sqrt{p}$ features per split, where p is the number of features. The input dataset is a set of data points for each pixel in our training dataset, consisting of the values of the imagery bands plus the categorical label for the corresponding classification. After training the classifier with these inputs, the trained classifier can then be applied to any image consisting of the same bands as the input training data. We applied each classifier to the Landsat composite image over the entire TIX, excluding the masked floodplain. The result of this process is a three-class categorical classification image over the TIX for each training dataset, at the same resolution as the input Landsat imagery (30 m). The random-forest classifier is a pixel-wise algorithm that does not consider spatial relationships between pixels; each pixel is assigned its class solely from its local spectral characteristics. As a result of the point-based nature of the classifier and the noise inherent to remote-sensing imagery, the resulting classification image exhibits pixel-scale noise that probably does not reflect true short-wavelength features. To reduce this noise and highlight the larger-scale features, we filtered the raw image to produce a filtered classification image. At each pixel, we applied a focal mode filter, taking the mode of all pixels within a circular window of radius 2 pixels (60 m). The final result of this filtering step is a three-class categorical classification image over the TIX for each training dataset, with pixel-scale noise removed.

Accuracy assessment

To assess the accuracy of the classification and filtering algorithm, we used a partitioning method in which we randomly divided the training data into a training set, which we input to train the classifier, and a validation set that we withheld from training, which we used to measure the success of the classifier in reproducing our assigned categorical

labels on areas it was not trained on. We divided the polygons by using a random number generator to assign a number between 0 and 1 to each polygon: polygons whose random number was less than 0.8 were assigned to the training set, and the remainder compose the validation set, such that on average 80% of polygons were used for training and 20% for validation. Given the small number of polygons in our training dataset compared with typical machine-learning applications, we iterated this process 100 times, using a different random split to train and validate the classifier in each iteration. Each iteration produced an accuracy matrix comparing the assigned class from the training data with the filtered output classification map; we summed the 100 resulting accuracy matrices to yield a cumulative accuracy matrix. To compute the two final classification maps, we used all training polygons to train each classifier, without partitioning the datasets. We performed this procedure separately for the two classifiers (secondary forest and undisturbed forest).

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

AIKAX owns the intellectual property rights (copyright) to all digital data, images and maps created or extrapolated from this project. These data are housed in the AIKAX-maintained ArcGIS Portal, for which there are established protocols for data sharing and collaboration. Requests for access should be made to mheck@ufl.edu.

Code availability

The Google Earth Engine script for image processing and classification is available via Zenodo (<https://doi.org/10.5281/zenodo.11039509>) (ref. 49).

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Acknowledgements

Collaborative field research in the TIX was authorized by the Associação Indígena Kuikuro do Alto Xingu (AIKAX) for NSF Archaeology grants sponsored by the University of Florida from 2002 to 2021 (NSF BCS#0004487, 0353129, 1660459) (M.H.) in collaboration with Brazilian institutional sponsors, the Museu Nacional, Museu Paraense Emílio Goeldi and Universidade de São Paulo. Authorizations were awarded to these institutions for the NSF-sponsored research from the Conselho Nacional de Pesquisa e Tecnologia (CNPq), the Fundação Nacional do Índio (FUNAI), Instituto de Patrimônio Histórico e Artística Nacional (IPHAN), Conselho de Ética (CONEP) and Conselho Nacional de Arqueologia (CNA) in Brazil. Additional support was provided by the University of Florida, the Wenner–Gren Foundation and the São Paulo Research Foundation (grant 2017/25157-0 to J.W.), conducted under the auspices of the authorized NSF-sponsored project. The AIKAX Geospatial Portal is supported by the William Talbott Hillman Foundation, the Pennywise Foundation and Puente Institute. In addition to co-authors on this

paper, the Kuikuro community at large is gratefully acknowledged, including contributions as paid assistants by over three dozen community members over the past 30 years. The remote-sensing analyses presented in this paper are based specifically on work conducted at MIT by S.L.G., M.J.S., J.D.H. and J.T.P., supported by the National Aeronautics and Space Administration Earth and Space Science Fellowship under grant no. 80NSSC18K1324 (S.L.G.) and the MIT Abdul Latif Jameel Water and Food Systems Lab (J.T.P.). Research at MIT was also sponsored by the United States Air Force Research Laboratory and the Department of the Air Force Artificial Intelligence Accelerator and was accomplished under Cooperative Agreement Number FA8750-19-2-1000 (J.T.P.). The views and conclusions contained in this document are those of the authors and should not be interpreted as representing the official policies, either expressed or implied, of the Department of the Air Force or the US Government. The US Government is authorized to reproduce and distribute reprints for Government purposes notwithstanding any copyright notation herein.

Author contributions

S.L.G. performed image classification and analysis. S.L.G., M.J.S., M.H. and J.T.P. wrote the manuscript with input from all authors. S.L.G. and M.J.S. curated training data. S.L.G. and J.D.H. developed the code for image processing and classification. M.J.S., M.H., B.F., H.L., J.W., B.M., W.B.D., C.F., K.W., H.K. and T.W.K. contributed field mapping used in the creation of training data. M.H. directed NSF-sponsored archaeological research in the TIX. B.F. and M.H. acquired the necessary permits for research in Brazil. A.K. facilitated and provided oversight of field research in the Kuikuro village in the TIX.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41893-024-01399-3>.

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Peer review information *Nature Sustainability* thanks Michael Palace, Alessandro Samuel-Rosa and the other, anonymous, reviewer(s) for their contribution to the peer review of this work.

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Research sample	Our research samples are Landsat 5 satellite imagery and land-cover GIS polygons.
Sampling strategy	Sample sizes were based on available field data.
Data collection	Training polygons were drawn by authors Samuel Goldberg and Morgan Schmidt, as described in the methods.
Timing and spatial scale	We use satellite imagery from 1985-2005, over the 26000 km2 Território Indígena do Xingu
Data exclusions	We excluded the floodplain regions from analysis, since our classification focuses on differences in upland forest.
Reproducibility	We assessed the accuracy and reproducibility of our classification algorithm using a repeated random split between training and validation sets, as described in the methods.
Randomization	For accuracy assessment, we randomly assigned polygons to training and validation sets using a random number generator.
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Did the study involve field work?	☒ Yes ☐ No

Field work, collection and transport

Field conditions	Variable (fieldwork conducted over many years)
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